

Lecture 3 – Newton’s equation of motion

- A/Prof Tay Seng Chuan

Email: pvotaysc@nus.edu.sg

In this lecture we will consider Newton’s answer to this question,

“How do the position and velocity of a particle change in the course of time?”



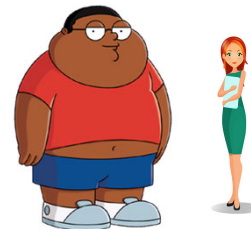
A few slides in this chapter come from Chapter-4 and Chapter-5 of Young’s textbook.

This PPT file will well define the scope of your tests.

1

Recall Newton’s First Law

- The Newton’s First Law can be understood in this simple phrase: “If the net external force applying on an object is 0, the object will continue with its previous state.” So if the object was not moving, it will not move now and later. If the object was moving with speed p m/s along the direction q , it will also be moving with speed p m/s along the direction q now and later. This is called the law of inertia – the tendency to do the same thing when the **vector sum** of all forces acting on the object is a zero vector.
- In this course we have to learn two types of inertia in mechanics: **translational inertia** and **rotational inertia**.
- The measurement of translational inertia (the reluctance to move along a straight line) is the mass of the object (in kg).
- The measurement of rotational inertia (the reluctance to rotate) is new to many of us and will be discuss in Lecture-7 and will be very interesting.



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Measurement of Inertia by Mass (in kg)



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Recall Newton's Third Law

- The Newton's Third Law is about the action-and-reaction pair when two objects are in contact.
- The contact may have physical touch, and is also valid if the two objects do not have any physical touch (e.g., the north pole of two magnets are approaching each other).
- One of the forces is the action force, the other is the reaction force.
- It doesn't matter which one is considered the action and which one is the reaction.
- The action force is equal in magnitude to the reaction force and opposite in direction.
- The action and reaction forces must act on the opposite objects.

Takeaway from the Newton's Third Law: A single isolated force cannot exist.
Action comes with reaction. This is the law of nature !!!

Common mistake made by students about the Newton's Third Law: Action can cancel reaction – *this is wrong*. Action cannot cancel reaction because they act on the opposite body !!!

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Newton's equation of motion

Consider a particle of mass m , moving on the xy -plane, subject to a force F . Newton's equation of motion relates the acceleration a of the particle to the force F (Newton's Second Law). :

$$F_x = ma_x, \quad F_y = ma_y. \quad (\text{Likewise for the } z \text{ axis. The subscripts indicate the relevant axis.})$$

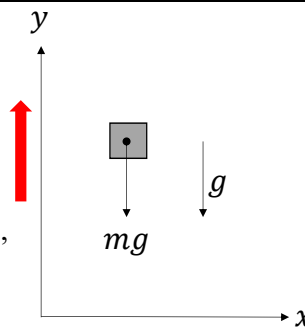
Remarks.

- The **(net or total) force** acting on the particle is **zero** if and only if the **acceleration** of the particle is **zero**, i.e., the particle is at rest or moving with constant velocity – **inertia (Newton's First Law)**.
Law of inertia!!
- Examples of force:
 - weight;
 - tension;
 - normal contact force;
 - static frictional force, kinetic frictional force;
 - fluid resistance force;
 - spring force.

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Example 1. Weight – a freely falling object

- Mass of the object is represented by m
- SI unit of mass is kg.
- Object is **falling freely** with **acceleration**, i.e.,
 $a_x = 0, a_y = -g$
- The object experiences a **force**, i.e., *Why negative!!*
 $F_x = 0, \quad F_y = -mg.$
- The **weight** of the object is always pointing downward. *Why again?*
- For example, if $m = 1 \text{ kg}$ then its **weight** has magnitude $1 \text{ kg} \times 9.80 \text{ m/s}^2 = 9.80 \text{ N}$, and the direction of weight is downward.
- Unit of force: 1 **newton** (N) = $1 \text{ kg} \times 1 \text{ m/s}^2$.



Mass (in kg) is a measurement of inertia, and weight (in N) is a measurement of force. They are not the same!!

This example will be a weightless scenario if air resistance is negligible. Weightlessness does not mean that the object has no weight.

Weightlessness is just a sensation of not being touched.

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Example 2. Tension – an object with a string attached

- T is the **tension** in the string.
- Mass of the object is m
- Object is in **equilibrium**, i.e.,
 $a_x = 0, a_y = 0$
- The object experiences **zero net force**,
 $F_x = 0, F_y = T - mg = 0.$

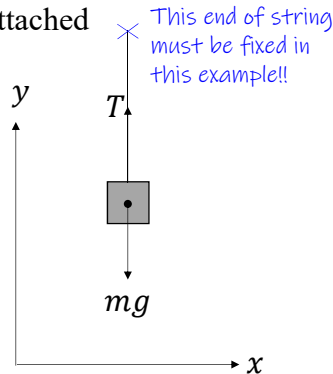
- For example, if $m = 3$ kg then

$$T = mg = (3 \times 9.80) \text{ N}$$

$$= 29.4 \text{ N}$$

What do we mean by $T = mg$?
Answer:

It means tension and weight have the same magnitude because there is no acceleration, but they point in opposite direction to each other.



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Example 2. Tension – an object with a string attached (cont.)

- T is the **tension** in the string.
- Mass of the object is m
- Object is moving in the positive y-direction with **acceleration** $a_y \neq 0$, and with $a_x = 0$
- The object experiences a **net force**,
 $F_x = 0, F_y = T - mg = ma_y.$

- F_y is the net force on the object.

- For example, if $m = 1$ kg and $a_y = 5.65 \text{ m/s}^2$, then

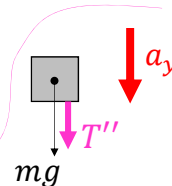
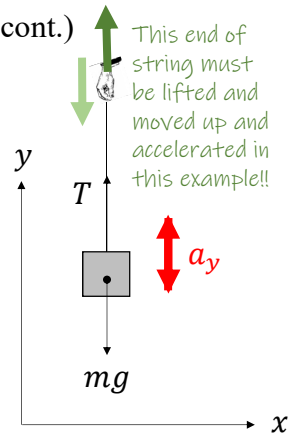
$$T = mg + ma_y = 1 \times (9.8 + 5.56) = 15.36 \text{ N}.$$

Question.

Why the magnitude of tension is larger than magnitude of the weight?

What would the tension be if the object were moving in the negative y-direction with acceleration $a_y = -5.65 \text{ m/s}^2$? $\rightarrow mg - T' = ma_y \rightarrow T' = mg - ma_y = 1(9.8 - 5.65) = 4.15 \text{ N} < T$

What if $a_y = -14.8 \text{ m/s}^2$? $\rightarrow mg - T'' = ma_y \rightarrow T'' = mg - ma_y = 1(9.8 - 14.8) = -5 \text{ N}$



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Example 2. Tension – an object with a string attached (cont.)

- T is the **tension** in the string and is not vertical.
- Mass of the object is m
- Object is moving in the positive x -direction
with **acceleration** $a_x \neq 0$, and $a_y = 0$
- The object experiences a **net force**,

$$F_x = T \sin \theta = ma_x$$

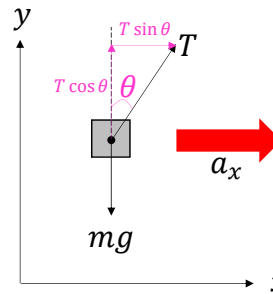
$$F_y = T \cos \theta - mg = 0 \rightarrow T \cos \theta = mg$$

- For example, if $m = 1 \text{ kg}$ and $a_x = 5.65 \text{ m/s}^2$,
then

$$T \sin \theta = 1 \times 5.65, \quad T \cos \theta = 1 \times 9.80$$

$$\text{So, } \tan \theta = \frac{5.65}{9.80} \rightarrow \theta = 29.96^\circ \rightarrow T \cos 29.96^\circ = 9.80$$

$$T = 11.31 \text{ N}$$



Linear motion along the x-axis only!!

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Example 2. Tension – an object with a string attached (cont.)

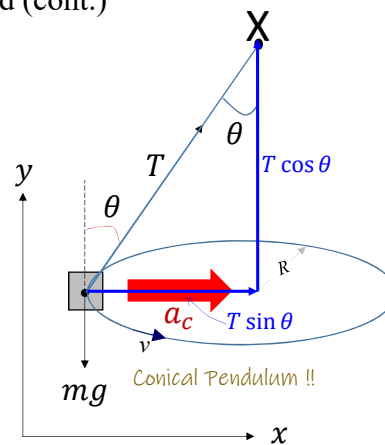
- Object is moving in the positive x -direction
with **acceleration** $a_y = 0$, and $a_x \neq 0$
- T is the **tension** in the string and is not vertical.
- The other end of string is fixed at X.
- Mass of the object is m
- Object will be moving on a horizontal circle of radius R with constant speed v , and so with $a_y = 0$, and
centripetal acceleration $a_c = \frac{v^2}{R}$.

- The object experiences a **net force**,

$$\text{Horizontal Force} = T \sin \theta = ma_c = m \left(\frac{v^2}{R} \right)$$

$$\text{Vertical Force} = T \cos \theta - mg = 0 \rightarrow T \cos \theta = mg$$

$$\tan \theta = \frac{v^2}{Rg}$$

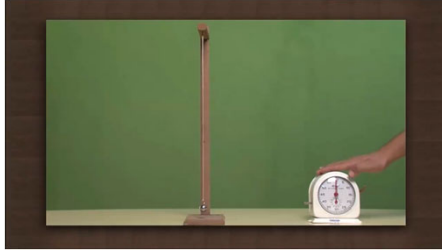


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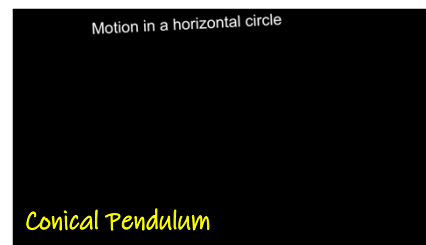
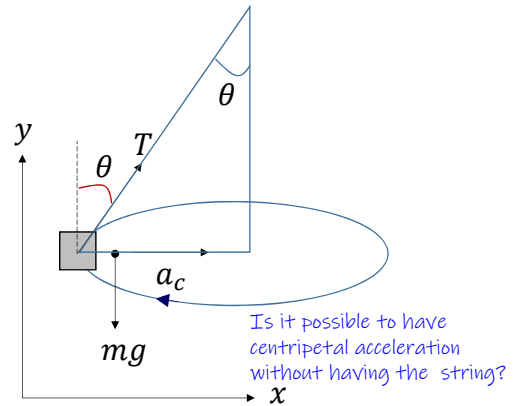
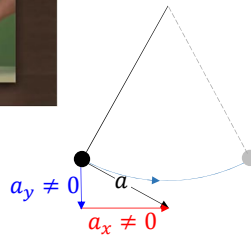
Example 2. Tension – an object with a string attached (cont.)

- Object is moving in the positive x -direction with **acceleration** $a_x \neq 0$, and with $a_y = 0$
- The other end of string is fixed. $\tan \theta = \frac{v^2}{gR}$ Conical Pendulum

Explain why the setting is **not** for a simple pendulum.

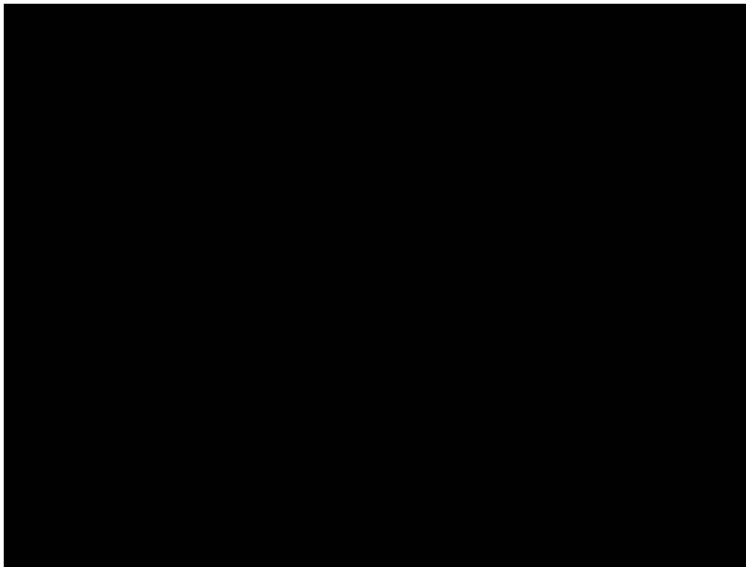


Simple Pendulum



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Centripetal Acceleration without a String



The centripetal force is provided by the static friction on the ground!!

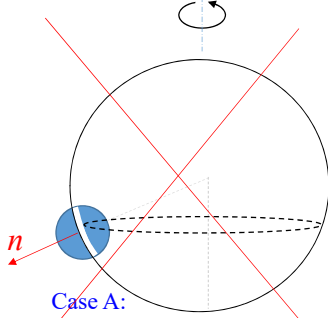
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Problem. A small bead can slide without friction on a circular hoop that is in a vertical plane and has a radius of 0.1 m as shown in the figure. The hoop rotates at a constant rate of 4.8 rev/s about a vertical diameter.

(i) Find the angle β at which the bead is in vertical equilibrium.

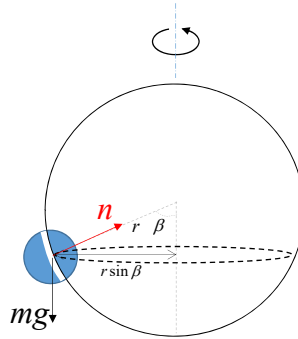
Answer:

First, we have to decide the direction of normal force (Case A or Case B?).



Case A:

Normal force is radially outward
Case A is not possible because the fictitious centrifugal force will push the bead further outward.



Case B: Normal force is radially inward
inertia frame

$$\text{Horizontally: } n \sin \beta = m \left(\frac{v^2}{r \sin \beta} \right) \dots (1)$$

$$\text{Vertically: } n \cos \beta = mg \dots (2)$$

linear velocity = frequency $\times$ circumference

$$\frac{(1)}{(2)}: \frac{\sin^2 \beta}{\cos \beta} = \frac{v^2}{rg} = \frac{(4.8 \times (2\pi(r \sin \beta)))^2}{rg}$$

$$\Rightarrow \frac{1}{\cos \beta} = \frac{4.8^2 (2\pi)^2 \times r}{g} = \frac{4.8^2 (2\pi)^2 \times 0.1}{9.8}$$

$$= 9.2815 \Rightarrow \beta = 83.8^\circ \text{ (the answer when observed from an inertia frame)}$$

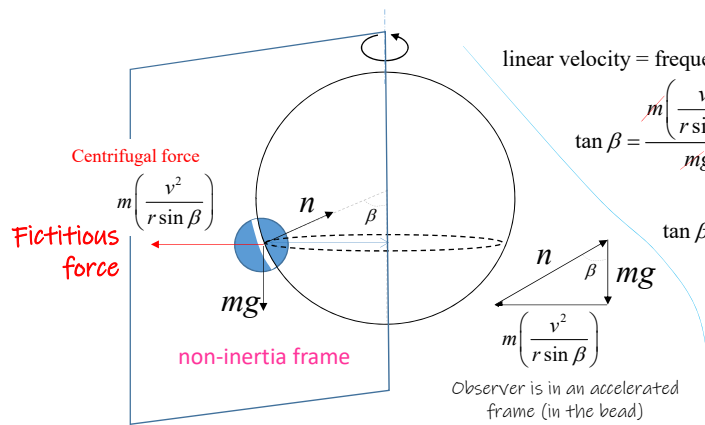
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Problem. A small bead can slide without friction on a circular hoop that is in a vertical plane and has a radius of 0.1 m as shown in the figure. The hoop rotates at a constant rate of 4.8 rev/s about a vertical diameter.

(i) Find the angle β at which the bead is in vertical equilibrium.

Alternate Approach:

A more straightforward way to solve this portion is based on the equilibrium of the bead on the rotating vertical plane.



linear velocity = frequency $\times$ circumference

$$\tan \beta = \frac{m \left(\frac{v^2}{r \sin \beta} \right)}{mg} = \frac{v^2}{gr \sin \beta} = \frac{(4.8 \times (2\pi(r \sin \beta)))^2}{gr \sin \beta}$$

$$\tan \beta = \frac{\sin \beta}{\cos \beta} = \frac{4.8^2 (2\pi)^2 r \sin^2 \beta}{gr \sin \beta}$$

$$\frac{1}{\cos \beta} = \frac{4.8^2 (2\pi)^2 \times 0.1}{9.8} = 9.2815$$

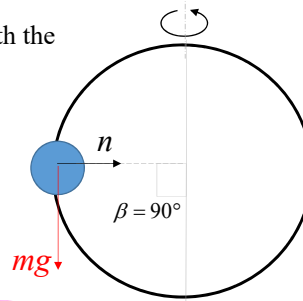
$$\Rightarrow \beta = 83.8^\circ \text{ (same answer when observed from a non-inertia frame)}$$

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- (ii) Is it possible for the bead to “ride” at the same elevation with the centre of the hoop? Explain.

Answer:

At the same elevation as the center of the hoop ($\beta = 90^\circ$), there is no force that can balance the bead's weight. As a result, the acceleration in the y -direction can never be zero. So it is **impossible** to keep the bead at the same elevation with the centre of the hoop.



- (iii) What will happen if the hoop rotates at 1.00 rev/s?

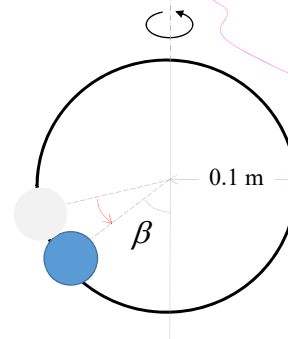
Answer:

$$\text{From } \frac{1}{\cos \beta} = \frac{(4.8 \text{ rev/s})^2 (2\pi)^2 \times 0.1}{9.8}, \quad \beta = 83.8^\circ.$$

If the hoop rotates at 1.00 rev/s from 4.8 rev/s, $\frac{1}{\cos \beta}$ will

yield a smaller value, implying a smaller β and causing the bead to go down to:

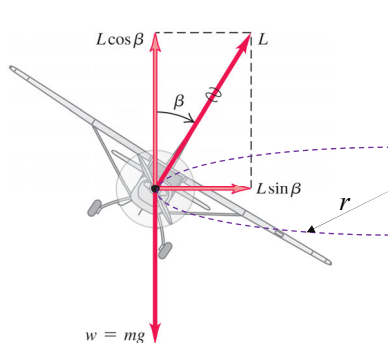
$$\frac{1}{\cos \beta} = \frac{(1 \text{ rev/s})^2 (2\pi)^2 \times 0.1}{9.8} \rightarrow \beta = \boxed{76.67^\circ}.$$



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Flight of Airplanes

- When an airplane is flying in a straight line at constant speed and at a steady altitude, the airplane's weight is exactly balanced by the lift force provided by the air mass.
- To make the airplane turn, the pilot banks the airplane one side so that the lift force has a horizontal component to act as the centripetal force. The vertical component of the lift continues to balance the weight.



$$\text{Horizontal forces: } L \sin \beta = m \left(\frac{v^2}{r} \right) \quad \dots (1)$$

$$\text{Vertical forces: } L \cos \beta = mg \quad \dots (2)$$

$$\frac{(1)}{(2)} : \tan \beta = \frac{v^2}{rg}$$

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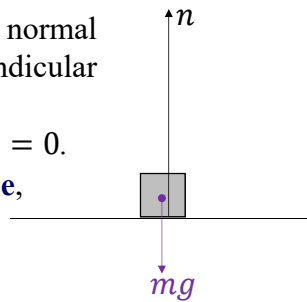
Example 3. Normal force – an object on a horizontal plane

- Mass of the object is m
- n is the **normal force**. The direction of normal force is from the contact surface perpendicular to the object.
- Object is in **equilibrium**, i.e., $a_x = 0$, $a_y = 0$.

So the object experiences a **zero net force**,

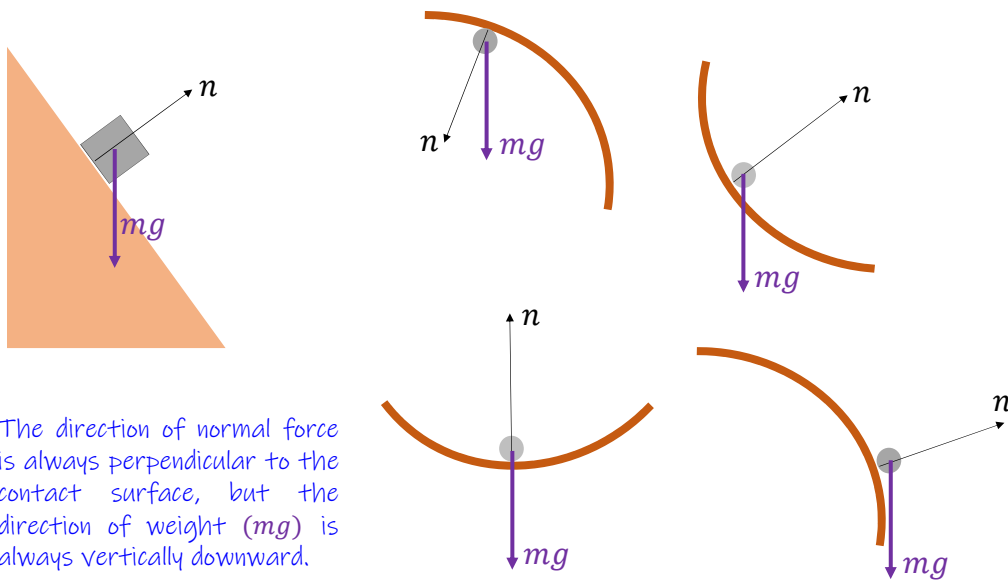
$$F_x = 0, \quad F_y = n - mg = 0.$$

- For example, if $m = 1$ kg then
 $n = mg = 1 \times 9.8 = 9.80$ N



In this case the contact surface must be a rigid body!!

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The direction of normal force is always perpendicular to the contact surface, but the direction of weight (mg) is always vertically downward.

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Example 3. Normal force – an object on a horizontal plane (cont.)

- Mass of the object m
- n is the **normal force**.
- Object is moving in the positive y -direction with **acceleration** a_y , and with $a_x = 0$
- The object experiences a **net force**,

$$F_x = 0, \quad F_y = n - mg = ma_y.$$

- For example, if $m = 1 \text{ kg}$ and $a_y = 5.65 \text{ m/s}^2$, then $n = mg + ma_y = 15.5 \text{ N}$. *There must be some mechanism in the surface, such as a spring to shoot up the object!!*

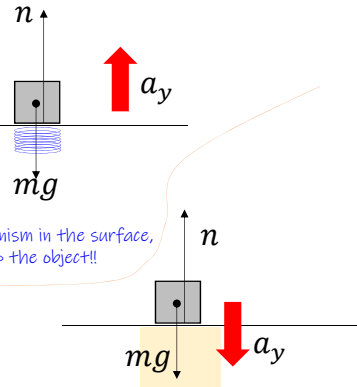
Question.

What would n be if the object were moving in the negative y -direction with acceleration $a_y = -4.8 \text{ m/s}^2$?

Answer:

In this case the contact surface cannot be a rigid body. The contact surface is soft!!

Take going down as positive:
 $mg - n = m \times 4.8 \Rightarrow n = m \times (9.8 - 4.8) = 5 \text{ N} \uparrow$
Normal force is still pointing upward but its magnitude is smaller than the weight!!



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Example 3. Normal contact force – an object on an inclined plane (cont.)

- Mass of the object is m
- n is the **normal force**.
- Object is moving in the positive **slanted** x -direction with **acceleration** $a_x \neq 0$, and with **slanted** $a_y = 0$
- The object experiences a **net force**,

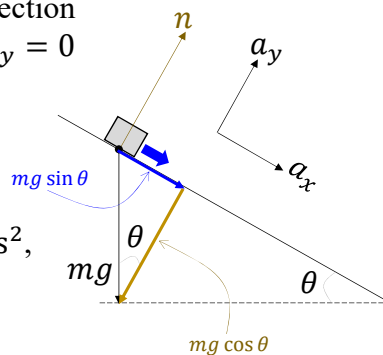
$$F_x = mg \sin \theta = ma_x$$

$$F_y = n - mg \cos \theta = 0$$

- For example, if $m = 1 \text{ kg}$ and $a_x = 4.90 \text{ m/s}^2$, then $\sin \theta = \frac{a_x}{g} = \frac{4.9}{9.8} = 0.5 \Rightarrow \theta = 30.0^\circ$

$$\Rightarrow n = mg \cos \theta = 1 \times 9.8 \times \cos 30.0^\circ = 8.49 \text{ N}.$$

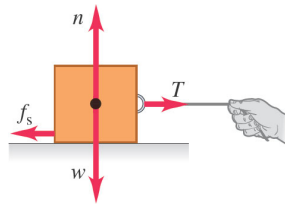
Why is the magnitude of normal force (8.49 N) smaller than the magnitude of the weight (9.8 N) ??



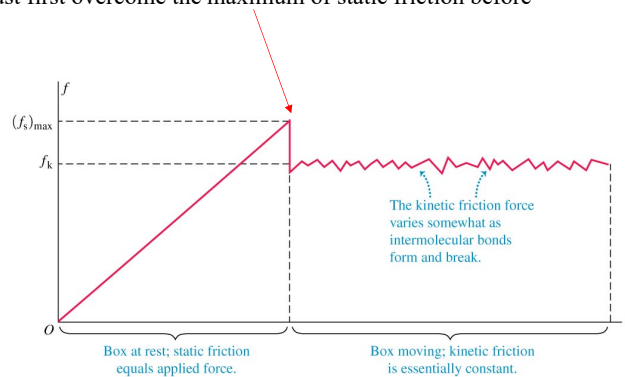
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Static Friction Followed by Kinetic Friction

- Before the box slides, static friction acts. But once it starts to slide, kinetic friction acts.
- Before the impending motion is taking place, the static friction is a variable when the applied force is increasing.
- The applied force must first overcome the maximum of static friction before the object can move.



② Weak applied force,
box remains at rest.
Static friction:
 $f_s < \mu_s n$



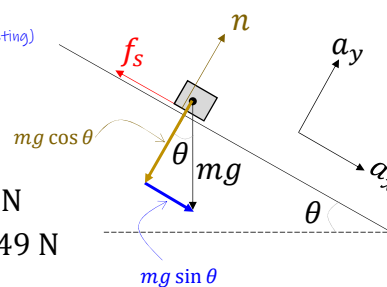
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Example 4. Static frictional force – an object resting (velocity = 0) on a rough inclined plane

- Mass of the object, m
- Object is in **equilibrium**, i.e., $a_x = 0$, $a_y = 0$
- f_s is the **static frictional force**.
- The object experiences a **zero net force** as it is resting, $F_x = mg \sin \theta - f_s = 0$ (Object is resting)
 $F_y = n - mg \cos \theta = 0$
- For example, if $m = 1 \text{ kg}$ and $\theta = 30.0^\circ$,
then $f_s = mg \sin \theta = 1 \times 9.8 \times 0.5 = 4.90 \text{ N}$
 $n = mg \cos \theta = 1 \times 9.8 \times 0.866 = 8.49 \text{ N}$

μ_s : coefficient of static friction
 f_s : static friction
$$\mu_s = \frac{f_s}{n}$$

Static friction $f_s = \mu_s \times n$



Question.

What could happen if θ is decreased or increased?

(please read next slide!!)

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Example 4. Static frictional force – an object resting (velocity = 0) on a rough inclined plane (cont.)

- For $\theta > \theta_{\max}$, object will start to slide. $\mu_s = \frac{f_s}{n}$
- $f_{s,\max}$ is the **maximum static frictional force**.
- At $\theta = \theta_{\max}$,

$$F_x = mg \sin \theta_{\max} - f_{s,\max} = 0,$$

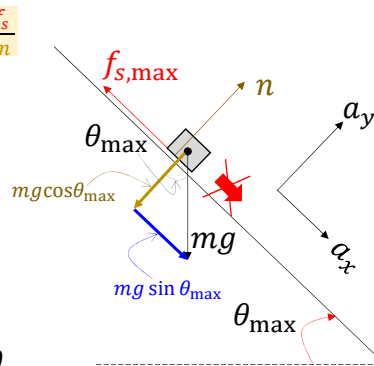
$$F_y = n - mg \cos \theta_{\max} = 0.$$

$$\rightarrow \left. \begin{array}{l} f_{s,\max} = mg \sin \theta_{\max} \\ n = mg \cos \theta_{\max} \end{array} \right\} f_{s,\max} = n \tan \theta_{\max}$$

$\theta_{\max}$ depends on the quality of the contact surface.

$$\tan \theta_{\max} = \mu_s \text{ (the coefficient of static friction).}$$

Static friction will be increasing when the angle of elevation is increased until impending motion starts!!
 When the object starts to slide, it experiences kinetic friction (i.e., no more static friction).



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Example 5. Kinetic frictional force – an object on an inclined plane

- Mass of the object is m
- Object is moving in the positive slanted x -direction with **acceleration** $a_x \neq 0$, and with slanted $a_y = 0$
- The object experiences a **net force**,

$$\begin{aligned} F_x &= mg \sin \theta - f_k = ma_x, \\ F_y &= n - mg \cos \theta = 0. \end{aligned}$$

- f_k is the **kinetic frictional force**:

$$f_k = \mu_k n$$

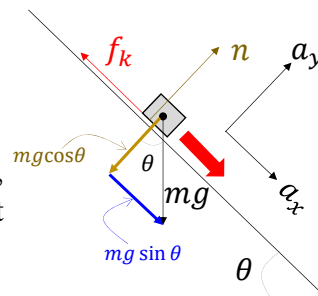
where μ_k is **coefficient of kinetic friction**, which depends on the quality of the contact surface. So, $a_x = g(\sin \theta - \mu_k \cos \theta)$.

- For example, if $\theta = 60.0^\circ$ and $\mu_k = 0.250$,
 then $a_x = 9.8(\sin 60.0^\circ - 0.25 \cos 60.0^\circ) = 7.26 \text{ m/s}^2$

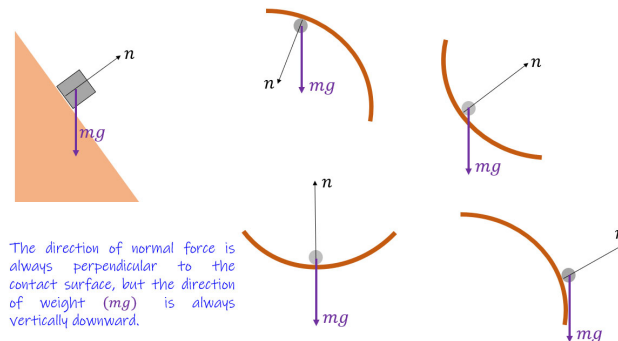
θ will have to be decreased in order to maintain a constant sliding velocity.

$$\text{Kinetic friction } f_k = \mu_k \times n$$

μ_k : coefficient of kinetic friction



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Important Note:

Kinetic friction is always equal to $\mu_k \times (\text{normal force})$.

But the static friction may not always equal to $\mu_s \times (\text{normal force})$.

The static friction can increase and can decrease depending on the applied force, and its **largest value** is $\mu_s \times (\text{normal force})$.

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Example 6. Static frictional force – an object on a rough inclined plane

- Mass of the object is m
- Object is moving on a horizontal circle of radius R with constant speed v , and so with $a_y \uparrow = 0$, and

centripetal acceleration $a_c = \frac{v^2}{R}$.

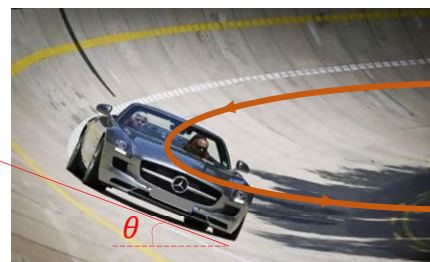
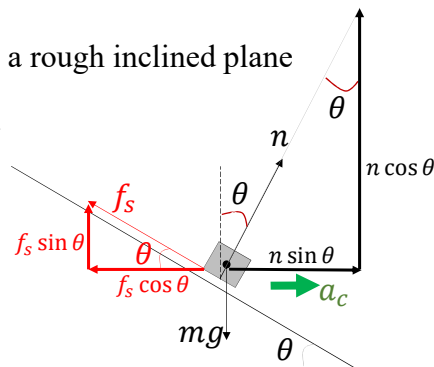
- n is the **normal contact force**.
- f_s is the **static frictional force**.

- The object experiences two components of **net force**, i.e.,

$$\text{Horizontally: } F_c = n \sin \theta - f_s \cos \theta = ma_c = m \left(\frac{v^2}{R} \right)$$

$$\text{Vertically: } F_y = n \cos \theta + f_s \sin \theta - mg = 0$$

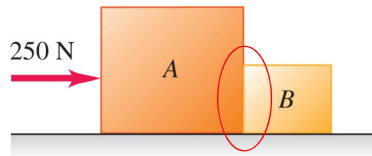
Why is f_k not used in the formulation?



Race track is a banked curve

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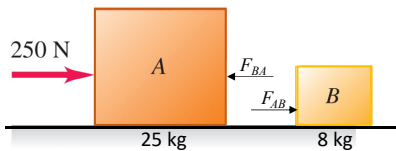
Problem on Force Cushion. Boxes A and B are in contact on a horizontal, frictionless surface. Box A has mass 25.0 kg and box B has mass 8.0 kg. A horizontal force of 250 N is exerted on box A. What is the magnitude of the force that box A exerts on Box B?



Answer:

A good guess will be the same 250 N but this is wrong.

Let F_{AB} be the force that box A exerts on Box B. Since the 250 N is to accelerate both boxes A and B, but F_{AB} is to accelerate only box B, we conclude that F_{AB} should be less than 250 N.



On Boxes A: $250 - F_{BA} = 25a \quad \dots (1)$

On Boxes B: $F_{AB} = 8a \quad \dots (2)$

By Newton's Third Law, $|F_{AB}| = |F_{BA}|$. Sub this in (1)

$$\rightarrow 250 - 8a = 25a \rightarrow a = 7.57 \text{ m/s}^2 \rightarrow a = 7.57 \text{ m/s}^2 \rightarrow F_{AB} = 8 \times 7.57 = 60.60 \text{ N} < 250 \text{ N}$$

Or, if we treat boxes A and B as one composite object, then we can apply the Newton's Second Law as $250 = (25 + 8)a$, and treat F_{AB} and F_{BA} as a pair of internal forces. Once the acceleration a is calculated, $F_{AB} = m_B a = 8a$.

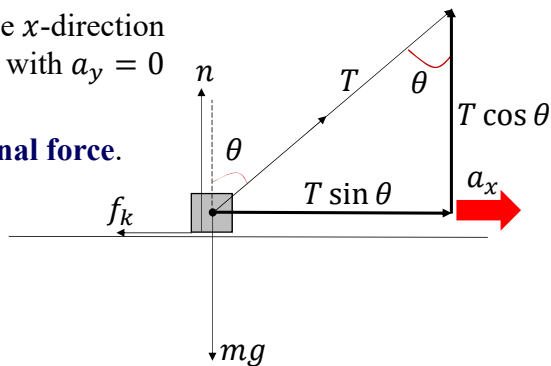
27

Example 7. An object on a rough horizontal plane, with a string attached

- Mass of the object is m
- Object is moving in the positive x -direction with **acceleration** $a_x \neq 0$, and with $a_y = 0$
- T is the **tension** in the string.
- $f_k = \mu_k n$ is the **kinetic frictional force**.
- n is the **normal contact force**.
- The object experiences a **net force**,

$$F_x = T \sin \theta - f_k \neq 0 = ma_x$$

$$F_y = n + T \cos \theta - mg = 0$$



$T \sin \theta > f_k$ for the linear motion to start (Yes or No?).

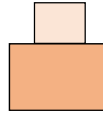
Answer:

No. It must be $T \sin \theta > \max(f_s)$ for the linear motion to start because the maximum of the static friction must be overcome first ($\mu_s > \mu_k$).

28

Problem. In a production site a 5.00-kg block is placed on top of a 12.0-kg block that rests on a frictionless table. The coefficient of static friction between the two blocks is 0.600. What is the maximum horizontal force that can be applied before the 5.00-kg block begins to slip relative to the 12.0-kg block,

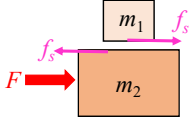
- if the force is applied to the more massive block, and
- if the force is applied to the less massive block ?



Answer:

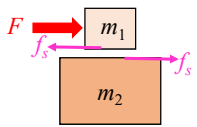
In order for sliding **not** to occur between the 2 blocks, the 2 blocks must have the same acceleration.

When the applied force F is increased, the static friction f_s will also be increased, and $f_s^{\max} = \mu_s m_1 g$.

(i) 

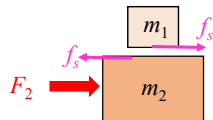
On m_1 : $f_s^{\max} = m_1 a \rightarrow \mu_s m_1 g = m_1 a \rightarrow a = \mu_s g$
 On m_2 : $F - f_s^{\max} = m_2 a \rightarrow F - \mu_s m_1 g = m_2 (\mu_s g)$
 $\rightarrow F = \mu_s g (m_1 + m_2) = 0.6 \times 9.8 (5 + 12) = 99.96 \text{ N}$

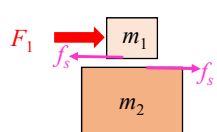
But what if the block on top has a larger mass?

(ii) 

On m_2 : $f_s^{\max} = m_2 a$
 On m_1 : $F - f_s^{\max} = m_1 a \rightarrow F - f_s^{\max} = m_1 \left(\frac{f_s^{\max}}{m_2} \right)$
 $\rightarrow F = f_s^{\max} \left(\frac{m_1}{m_2} + 1 \right) = \mu_s m_1 g \left(\frac{m_1}{m_2} + 1 \right) = 0.6 \times 5 \times 9.8 \left(\frac{5}{12} + 1 \right) = 41.65 \text{ N}$

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 $\rightarrow F_2 = \mu_s g (m_1 + m_2)$

 $\rightarrow F_1 = f_s^{\max} \left(\frac{m_1}{m_2} + 1 \right) = \mu_s m_1 g \left(\frac{m_1}{m_2} + 1 \right) = \mu_s g \left(m_1 \times \left(\frac{m_1}{m_2} \right) + m_1 \right)$

If we want to have a faster velocity, i.e., larger acceleration to move the 2 objects, is F_2 always greater than F_1 before the blocks slip?

Consider $F_2 - F_1$ and ignore $\mu_s g$.

$$\left(m_1 + m_2 \right) - \left(m_1 \times \left(\frac{m_1}{m_2} \right) + m_1 \right) = m_2 - m_1 \times \left(\frac{m_1}{m_2} \right) = m_2 \times \frac{m_2}{m_2} - m_1 \times \left(\frac{m_1}{m_2} \right) = \frac{m_2^2 - m_1^2}{m_2}.$$

If $m_2 > m_1$, as given in the question, we will push the lower block as $F_2 > F_1$ in order to have larger acceleration before the blocks slip.

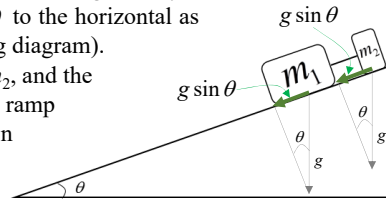
If $m_1 > m_2$, we can push the upper block in order to have larger acceleration before the blocks slip.

But we also have to consider the height of the center of mass (Lecture-6) of the 2 blocks to have a stable equilibrium. If the physical sizes of the blocks are the same, the advice is to place the more massive block at the bottom to lower the center of mass of the whole collection of objects.

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Problem. Two blocks made of different materials and connected together by a thin cord, slide down a plane ramp inclined at an angle θ to the horizontal as shown (block 2 is above block 1 as shown in the following diagram).

The masses of blocks 1 and 2 are m_1 and m_2 , when $m_1 > m_2$, and the coefficients of kinetic friction between the masses and the ramp are μ_1 and μ_2 respectively. Describe and explain the motion of the two blocks (i) if $\mu_1 < \mu_2$, (ii) if $\mu_1 > \mu_2$ once the two blocks are released from rest. (The μ here refers to the coefficient of kinetic friction.)



Answer:

Along the inclined plane there are two accelerations acting on m_1 and m_2 : acceleration due to gravity projected on the inclined plane, and, acceleration in the opposite direction (or called the deceleration) due the friction on the inclined plane. Both of these types of accelerations are **independent** of the masses of the blocks. **(The masses cancel each other on the both sides of equation!!)**

The acceleration experienced by both m_1 and m_2 due to gravity are equal, where the magnitude is $g \sin \theta$. Hence, the two blocks will travel at the same speed when friction is not taken into account. This acceleration **has nothing to do** with their masses.

- (i) If $\mu_1 < \mu_2$, m_1 will have smaller deceleration thus will slide with faster acceleration than m_2 (independent of the mass), and this results in the thin cord becoming taut (**tight with tension**). The taut cord exerts a tension on both blocks such that **the net acceleration on both blocks will finally become the same**.
- (ii) If $\mu_1 > \mu_2$, m_1 will have larger deceleration thus will slide with slower acceleration than m_2 (independent of the mass), and this results in the thin cord becoming slackened, and **m_2 will slide down to hit m_1** .

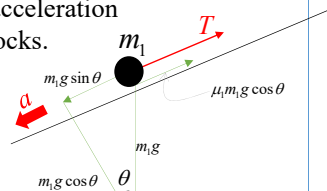
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Problem. Determine the expressions for the accelerations, a_1 and a_2 , of the blocks and the tension T in the cord for the two cases in the previous problem once the two blocks are released from rest. (a_1 is the acceleration for the lower block, a_2 is the acceleration for the higher block, and both accelerations are going downward and parallel to the inclined plane.) What is the implicit assumption made on the angle in the previous question? Why is the implicit assumption necessary?

Answer: The string will be tight!!

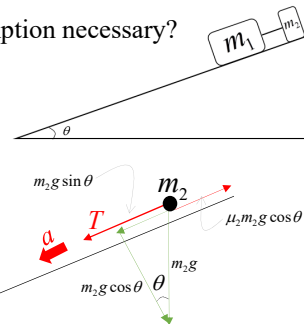
(i) $\mu_1 < \mu_2$:

Since acceleration of m_1 is the same as that for m_2 finally, we let a be the common acceleration of both blocks.



On m_1 ,

$$m_1 g \sin \theta - T - \mu_1 (m_1 g \cos \theta) = m_1 a \dots (1)$$



On m_2 ,

$$T + m_2 g \sin \theta - \mu_2 (m_2 g \cos \theta) = m_2 a \dots (2)$$

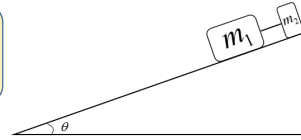
The rest are algebra!!!

32

(i) $\mu_1 < \mu_2$:

$$m_1 g \sin \theta - T - \mu_1 (m_1 g \cos \theta) = m_1 a \dots (1)$$

$$T + m_2 g \sin \theta - \mu_2 (m_2 g \cos \theta) = m_2 a \dots (2)$$



Add the two equations,

$$(m_1 + m_2)a = (g \sin \theta)(m_1 + m_2) - (g \cos \theta)(\mu_1 m_1 + \mu_2 m_2)$$

$$\rightarrow a = \frac{(g \sin \theta)(m_1 + m_2) - (g \cos \theta)(\mu_1 m_1 + \mu_2 m_2)}{m_1 + m_2}$$

$$= g \left(\sin \theta - \frac{(\mu_1 m_1 + \mu_2 m_2) \cos \theta}{m_1 + m_2} \right)$$

$$\text{So, } a_1 = a_2 = g \left(\sin \theta - \frac{(\mu_1 m_1 + \mu_2 m_2) \cos \theta}{m_1 + m_2} \right)$$

The string is tigh so the 2 blocks have the same acceleration!!

Now we can find tension T by substituting a into (2):

$$T = m_2 a - m_2 g \sin \theta + \mu_2 m_2 g \cos \theta$$

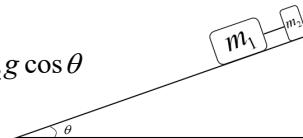
$$= m_2 g \left(\sin \theta - \frac{(\mu_1 m_1 + \mu_2 m_2) \cos \theta}{m_1 + m_2} \right) - m_2 g \sin \theta + \mu_2 m_2 g \cos \theta$$

These are algebra!!!

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(i) $\mu_1 < \mu_2$: Find the tension ...

$$T = m_2 g \left(\sin \theta - \frac{(\mu_1 m_1 + \mu_2 m_2) \cos \theta}{m_1 + m_2} \right) - m_2 g \sin \theta + \mu_2 m_2 g \cos \theta$$



$$\begin{aligned} \rightarrow (m_1 + m_2)T &= m_2 \left((m_1 + m_2) g \sin \theta - (\mu_1 m_1 + \mu_2 m_2) g \cos \theta \right) \\ &\quad - (m_1 + m_2) m_2 g \sin \theta + (m_1 + m_2) \mu_2 m_2 g \cos \theta \\ &= -(\mu_1 m_1 + \mu_2 m_2) m_2 g \cos \theta + (m_1 \mu_2 + m_2 \mu_2) m_2 g \cos \theta \\ &= (\mu_2 - \mu_1) m_1 m_2 g \cos \theta \end{aligned}$$

$$\rightarrow T = \frac{(\mu_2 - \mu_1) m_1 m_2 g \cos \theta}{m_1 + m_2}$$

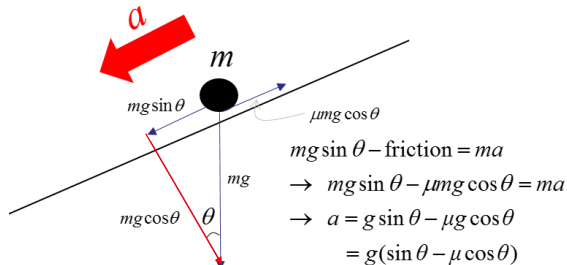
These are just algebra!!!

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(ii) $\mu_1 > \mu_2$:

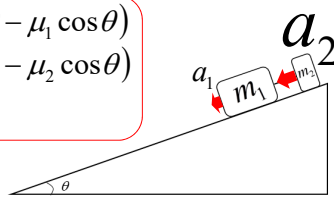
Since m_1 has more kinetic friction thus is slower, and the cord will be slacken, i.e., $T=0$.

$$\text{friction} = \mu \times (\text{normal force}) = \mu \times mg \cos \theta$$



So,

$$\begin{aligned}
 a_1 &= g(\sin \theta - \mu_1 \cos \theta) \\
 a_2 &= g(\sin \theta - \mu_2 \cos \theta) \\
 a_1 &< a_2
 \end{aligned}$$



The implicit assumption made on the angle θ is $\theta > \tan^{-1} \mu_s$, where μ_s is the maximum of $\mu_{s,1}$ (coefficient of static friction between block 1 and the inclined ramp surface), and, $\mu_{s,2}$ (coefficient of static friction between block 2 and the inclined ramp surface).

This assumption is to ensure that both objects 1 and 2 are able to overcome the static friction so that sliding is possible.

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Problem on Pulley System. In terms of m_1 , m_2 and g , find the acceleration of each block in the figure. There is no friction anywhere in the system.

Answer:

First we have to determine the relationship of the two accelerations a_1 (accelerate to the right) and a_2 (accelerate downward).

Consider the scenario where m_1 is displaced to the right by 1 m. In order for the system to remain taut (tight – without string slack), m_2 will have to descend. Since the moveable pulley is supported by the same string on its left and right, m_2 will not be lower by 1 m. Instead m_2 will be lower by 0.5 meter (so that 0.5 m (left) + 0.5m (right) = 1 m.)

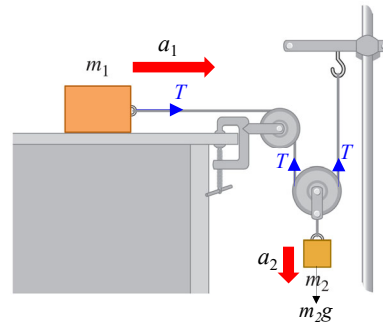
$$\text{On } m_1: T = m_1 a_1 \quad \dots (1)$$

$$\text{On } m_2: m_2 g - 2T = m_2 a_2 = m_2 \left(\frac{a_1}{2} \right) \quad \dots (2)$$

These are algebra!!

$$\text{Sub (1) to (2): } m_2 g - 2(m_1 a_1) = m_2 \left(\frac{a_1}{2} \right)$$

$$\rightarrow a_1 \left(2m_1 + \frac{m_2}{2} \right) = m_2 g$$



$$\rightarrow a_1 \left(\frac{4m_1 + m_2}{2} \right) = m_2 g$$

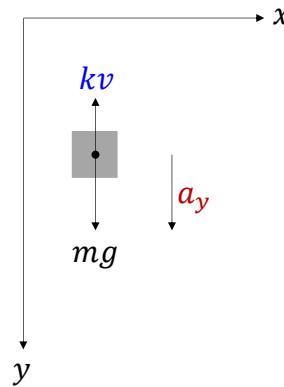
$$\text{On } m_1: a_1 = \frac{2m_2 g}{4m_1 + m_2}$$

$$\text{On } m_2: a_2 = \frac{a_1}{2} = \frac{m_2 g}{4m_1 + m_2}$$

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Example 8. An object falling in a fluid (Gas or Liquid)

- Mass of the object is m
- Object started from rest at $y = 0$ and is moving in the positive y -direction with **velocity** $v_x = 0$, $v_y = v$, **accelerations** $a_x = 0$, and $a_y = \frac{dv}{dt}$.
- The object experiences a **net force**,
 $F_x = 0$, $F_y = mg - kv = ma_y = m \frac{dv}{dt}$.
- kv is the **fluid resistance force**. Here, k is some positive constant.



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Example 8. An object falling in a fluid (Gas or Liquid) (cont.)

$$mg - kv = m \frac{dv}{dt}$$

- Divided by k and re-arranged, we have

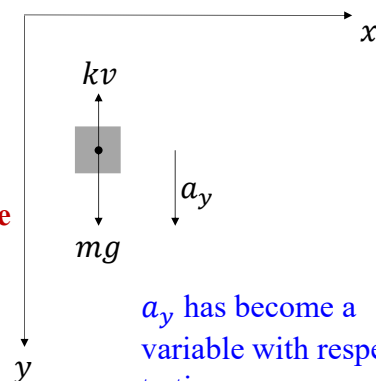
$$\frac{m}{k} \left(\frac{dv}{dt} \right) = \frac{mg}{k} - v.$$

- This is a **first-order variable separable differential equation**. We can write

$$\frac{dv}{\frac{mg}{k} - v} = \frac{k}{m} dt.$$

- Solve the equation by:

$$\int_0^v \frac{dv'}{\frac{mg}{k} - v'} = \int_0^t \frac{k}{m} dt'.$$



a_y has become a variable with respect to time so we **cannot** use the 4 kinematic formulae!!

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Example 8. An object falling in a fluid (Gas or Liquid) (cont.)

$$-\int_0^v \frac{dv'}{\frac{mg}{k} - v'} = \int_0^t \frac{k}{m} dt'$$

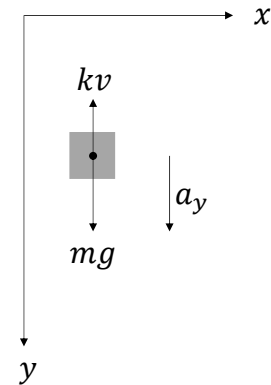
$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

$\log x$ is undefined if $x \leq 0$.

$$\log x - \log y = \log\left(\frac{x}{y}\right)$$

Integrating, we have

$$\begin{aligned} -\left[\ln\left|\frac{mg}{k} - v'\right|\right]_0^v &= \left[\frac{k}{m}t'\right]_0^t \\ \Rightarrow \ln\left|\frac{mg}{k} - v\right| - \ln\left|\frac{mg}{k} - 0\right| &= -\left(\frac{k}{m}t - 0\right) \\ \Rightarrow \ln\left|\frac{mg}{k} - v\right| - \ln\left|\frac{mg}{k}\right| &= -\frac{k}{m}t \\ \Rightarrow \ln\frac{\left|\frac{mg}{k} - v\right|}{\left|\frac{mg}{k}\right|} &= -\frac{k}{m}t \end{aligned}$$



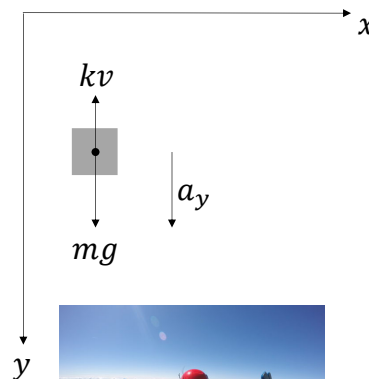
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Example 8. An object falling in a fluid (Gas or Liquid) (cont.)

$$\ln\frac{\left|\frac{mg}{k} - v\right|}{\left|\frac{mg}{k}\right|} = -\frac{k}{m}t$$

○ Recall $e^{\ln x} = x$. So $\frac{\left|\frac{mg}{k} - v\right|}{\left|\frac{mg}{k}\right|} = e^{-\frac{k}{m}t}$.

$$\begin{aligned} \text{If } v < \frac{mg}{k} \Rightarrow \frac{mg}{k} - v &= \frac{mg}{k} e^{-\frac{k}{m}t} \\ \Rightarrow v = v(t) &= \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t}\right) \end{aligned}$$



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Example 8. An object falling in a fluid (cont.)

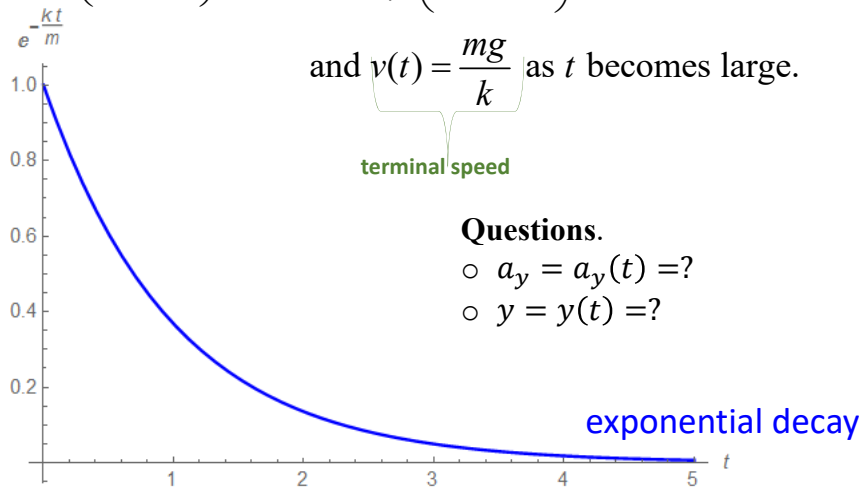
$$v(t) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right), \quad v(0) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m} \times 0} \right) = \frac{mg}{k} (1 - 1) = 0$$

$$\text{and } v(t) = \frac{mg}{k} \text{ as } t \text{ becomes large.}$$

terminal speed

Questions.

- $a_y = a_y(t) = ?$
- $y = y(t) = ?$



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Example 8. An object falling in a fluid (Gas or Liquid) (cont.)

$$v = v(t) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right)$$

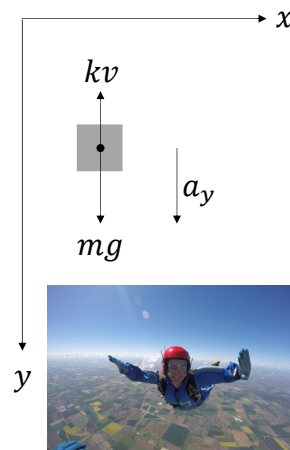
Let $m = 70$ kg, $k = 110$ kg/s for a certain parachute.

$$\begin{aligned} \Rightarrow v(t) &= \frac{70 \times 9.8}{110} \left(1 - e^{-\frac{110}{70}t} \right) \\ &= 6.2363 \left(1 - e^{-1.5714t} \right) \end{aligned}$$

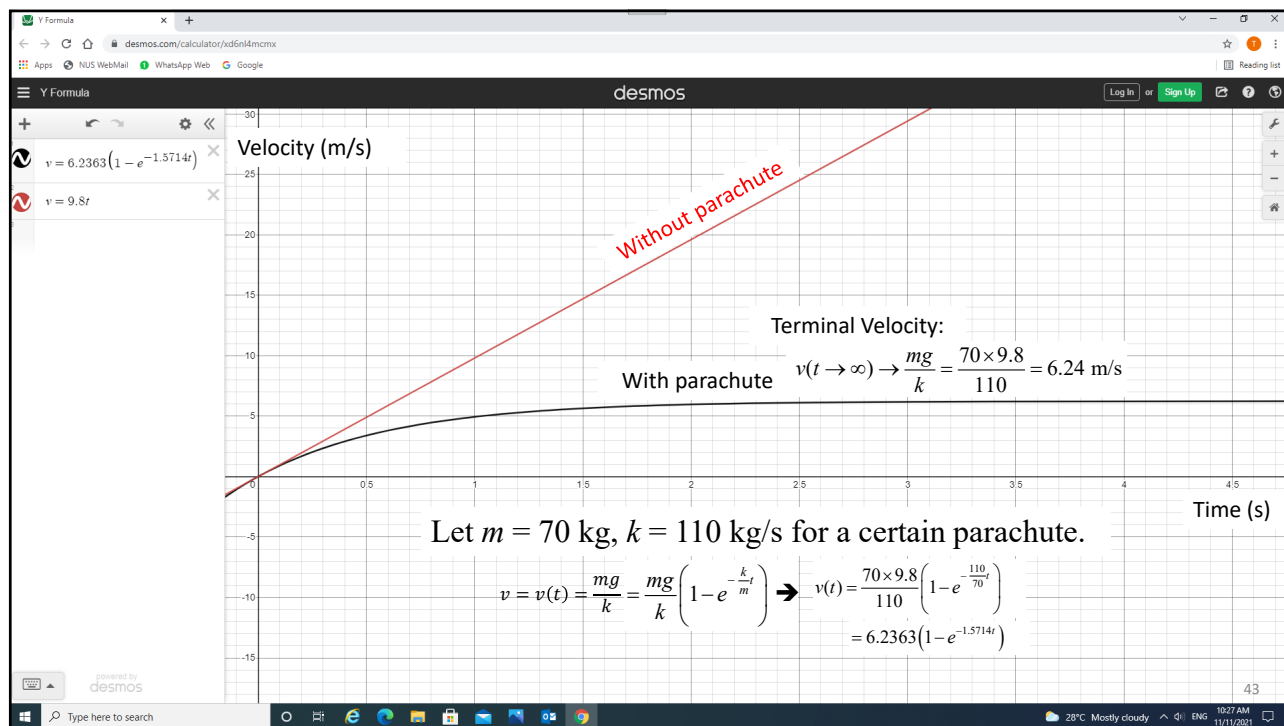
Without parachute:

$$v_f = v_i + at$$

$$\Rightarrow v_f = 9.8t$$



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Example 8. An object falling in a fluid (Gas or Liquid) (cont.)

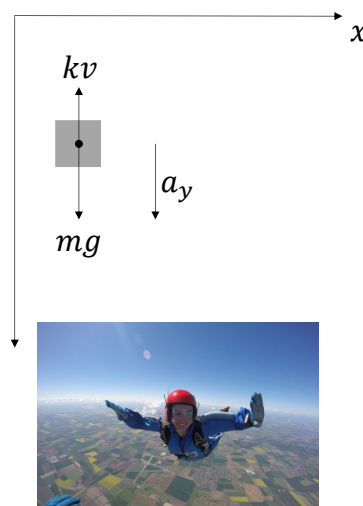
From $v = v(t) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right)$

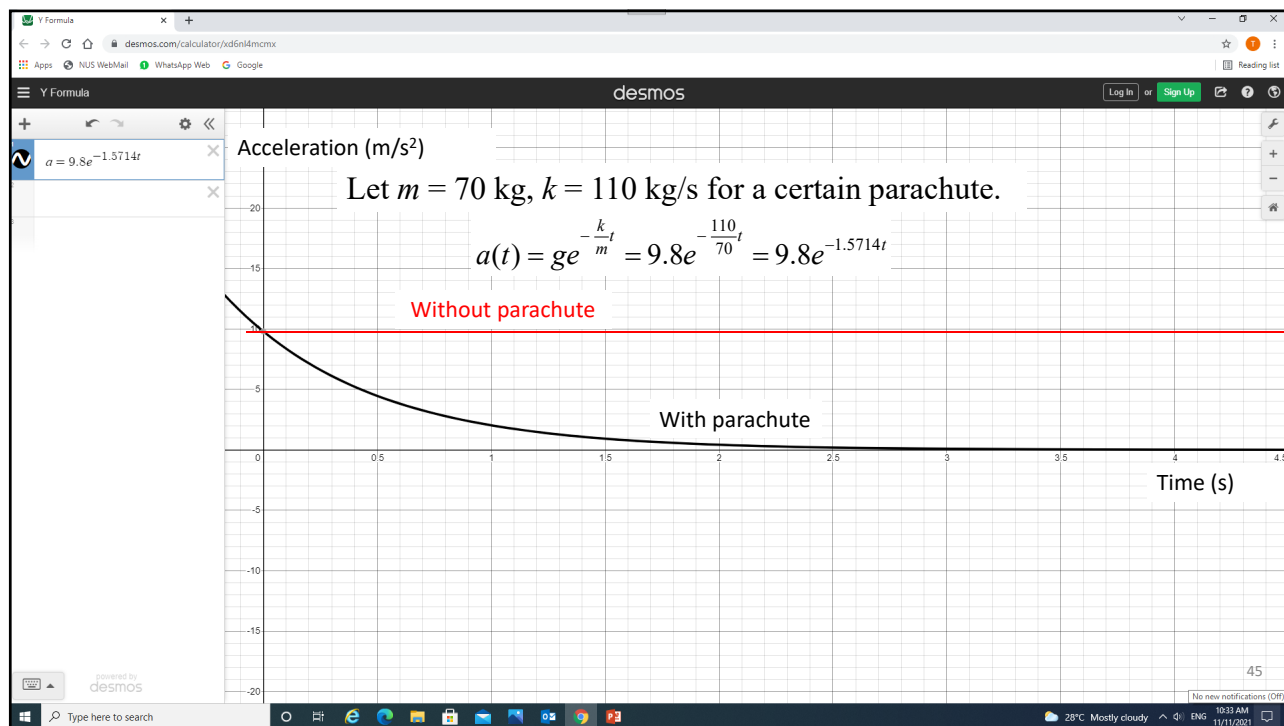
$$\rightarrow a(t) = \frac{d}{dt} \left(\frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right) \right)$$

$$= \frac{mg}{k} \left(0 - \left(-\frac{k}{m} \right) e^{-\frac{k}{m}t} \right)$$

$$= ge^{-\frac{k}{m}t} \rightarrow a(t=0) = g$$

$$a(t \rightarrow \infty) \rightarrow 0$$





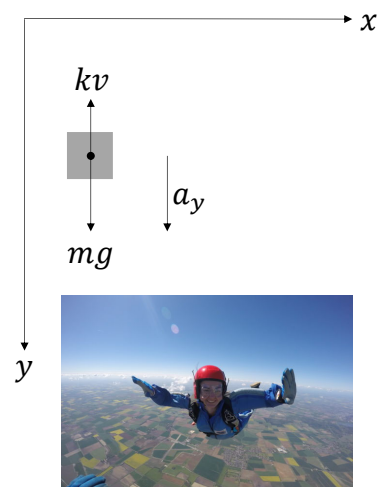
Example 8. An object falling in a fluid (Gas or Liquid) (cont.)

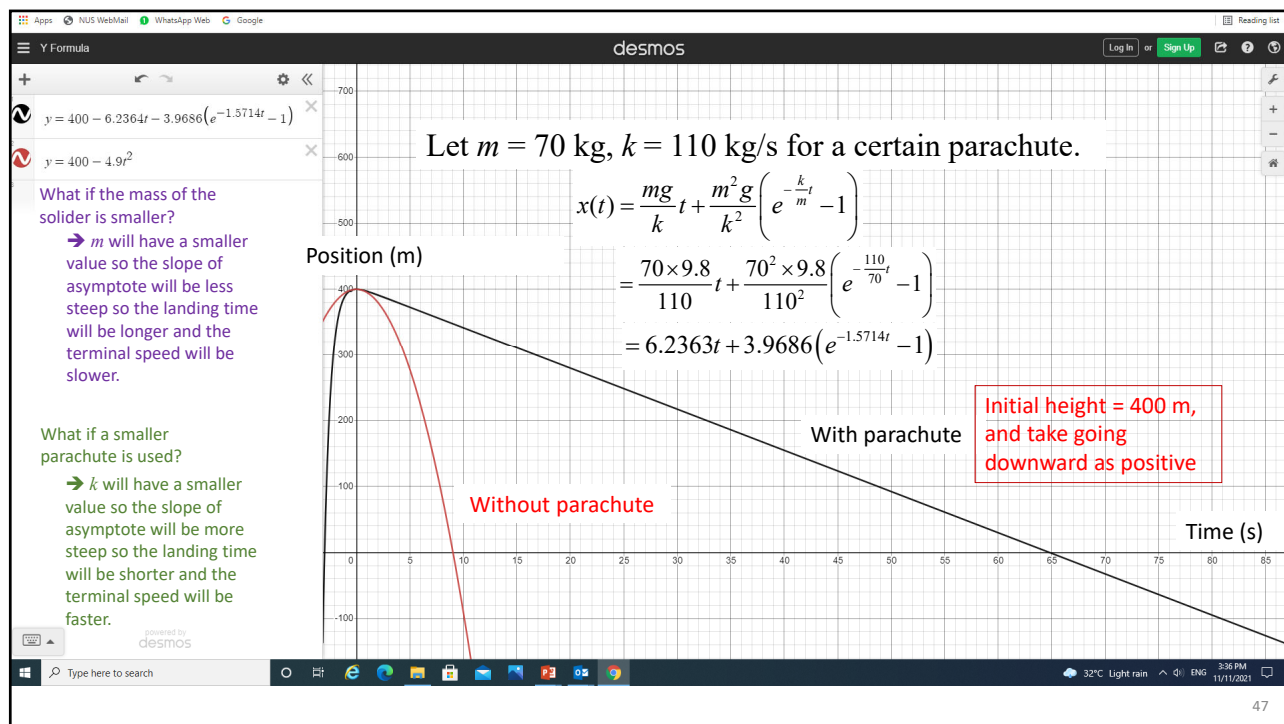
$$\text{From } v(t) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right) \rightarrow \frac{dx}{dt}(t) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right)$$

$$\rightarrow \int_0^x dx'(t) = \int_0^t \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t'} \right) dt' \rightarrow x(t) = \frac{mg}{k} \int_0^t \left(1 - e^{-\frac{k}{m}t'} \right) dt'$$

$$\begin{aligned} \rightarrow x(t) &= \frac{mg}{k} [t']_0^t - \frac{mg}{k} \int_0^t e^{-\frac{k}{m}t'} dt' \\ &= \frac{mg}{k} (t-0) - \frac{mg}{k} \int_0^t \left(-\frac{m}{k} \right) \left(\left(-\frac{k}{m} \right) e^{-\frac{k}{m}t'} \right) dt' \\ &= \frac{mg}{k} t + \frac{m^2 g}{k^2} \left[e^{-\frac{k}{m}t'} \right]_0^t = \frac{mg}{k} t + \left(\frac{m^2 g}{k^2} \left(e^{-\frac{k}{m}t} - 1 \right) \right) \end{aligned}$$

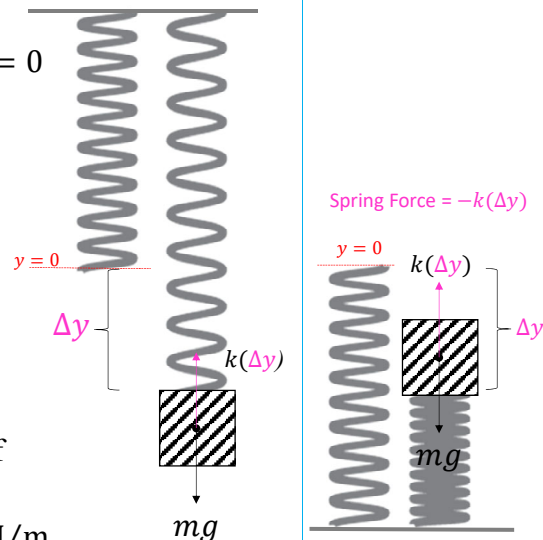
$$\rightarrow x(t \rightarrow \infty) = \frac{mg}{k} t + \left(\frac{m^2 g}{k^2} (0-1) \right) = \frac{mg}{k} t - \frac{m^2 g}{k^2} \quad (\text{linear})$$



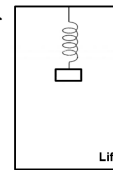


Example 9. Spring force – an object with a spring attached

- Mass of the object is m
- Object is in **equilibrium**, i.e., $a_x = 0$, $a_y = 0$
- The direction of spring force is always in the **opposite** direction of the deformation.
- The object experiences **zero net force**,
 $F_x = 0$, $F_y = mg - |-k(\Delta y)| = 0$.
- $k(\Delta y)$ is the magnitude of **spring force** – **Hooke's law**. Here, k is some positive constant – the **force** (or **spring**) **constant** of the spring, and Δy is the **elongation** of the stretched spring or the **compression** of the compressed spring.
- For example, if $m = 1$ kg and $k = 49.0$ N/m, then $|-k(\Delta y)| = mg = 9.80$ N $\rightarrow \Delta y = 0.200$ m.



Problem. A light spring is of natural length of 25.0 cm. It is suspended from the ceiling of a lift, and a mass is hung from the end of the spring. When the lift is moving downwards at constant speed, the length of the spring becomes 50.0 cm as shown in the figure.

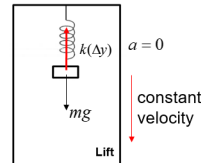


The lift then slows down with a constant deceleration of 2.0 m/s^2 . What will be the change in the length of the spring? Has the spring become longer or shorter?

Answer:

When the lift moves down at constant speed, **net** force = 0 N.

$$|-k(\Delta y)| = mg \rightarrow k = \frac{mg}{\Delta y} = \frac{m \times 9.8}{\underbrace{0.5 - 0.25}_{\text{elongation}}} = 39.2 \frac{\text{N}}{\text{m}}$$

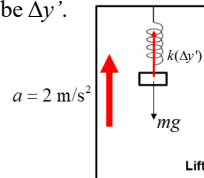


When lift decelerates, $\sum F = ma$. Let the new extension be $\Delta y'$.

$$|-k(\Delta y')| - mg = ma \rightarrow (39.2 \text{ m}) \Delta y' - m(9.8) = m(2)$$

$$\rightarrow \Delta y' = \frac{2 + 9.8}{39.2} = 0.3010 \text{ m (new elongation)}$$

$$\Delta y' - \Delta y = 0.3010 - 0.25 = 0.0510 \text{ m}$$



- Further stretched by 5.10 cm from the previous elongation. The spring has become longer from the previous elongation because the upper end of the spring is slowing down due to deceleration, but the mass at the lower end of the spring continues to go downward with its previous velocity due to its inertia.

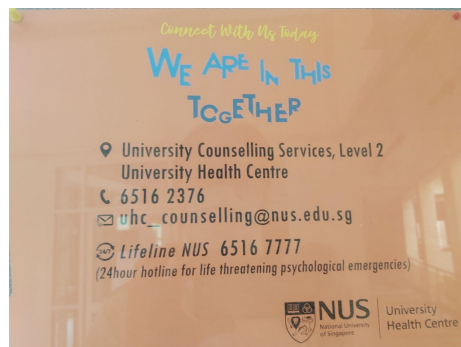
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Helplines to de-stress

We are not superman/supergirl. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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